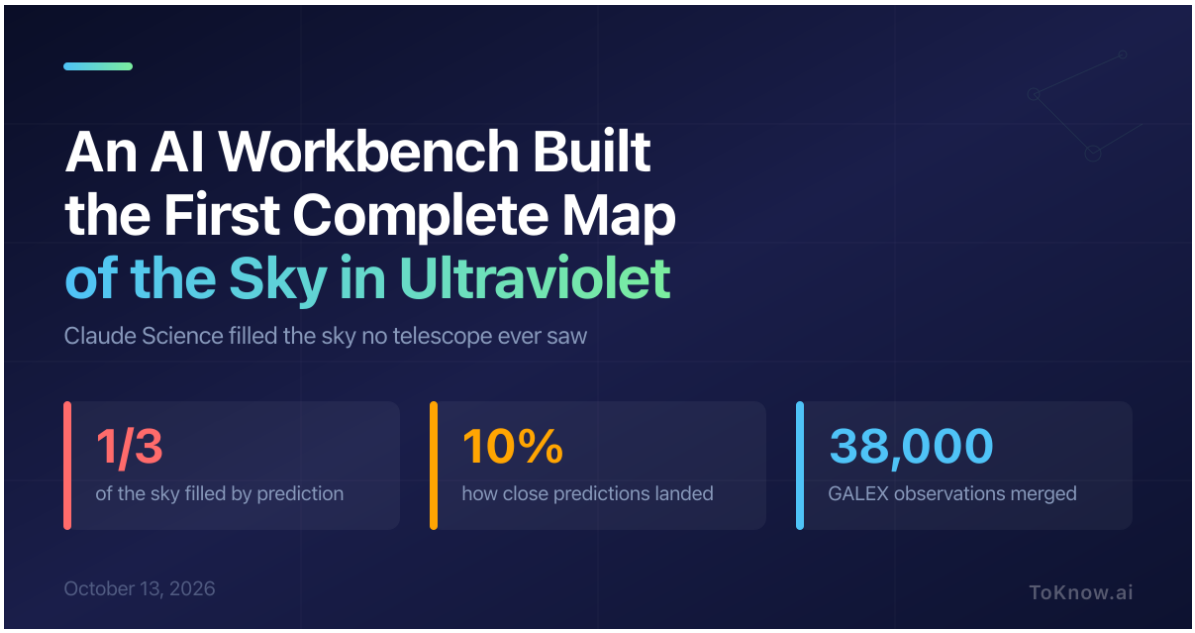
An AI Workbench Built the First Complete Map of the Sky in Ultraviolet

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An astrophysicist has built the first complete map of the sky in ultraviolet light, filling the roughly one third that no UV telescope has ever photographed. Brice Ménard, who works at Johns Hopkins University and at Anthropic, made it with Claude Science, Anthropic's workbench for scientists, which runs a lead agent that spins up helper agents and a reviewer that checks their work. Ultraviolet light is blocked by the ozone layer, so it can only be

measured from space. NASA's GALEX telescope imaged about two thirds of the sky in 38,000 visits between 2003 and 2013, but skipped the brightest regions, including most of the Milky Way's plane, to protect its detectors. Ménard's map merges GALEX with Swift, FIMS/SPEAR, TD-1, Planck and Gaia.

To fill the gaps, Claude used inpainting, where a model learns how the parts of an image relate and estimates what is missing. It learned how UV brightness tracks other wavelengths across the measured two thirds, then applied that to the unobserved third. Tested by hiding real UV patches, its guesses landed within about 10% of the actual measurements, a difference the eye cannot see. The finished map labels every pixel as measured or predicted, with an uncertainty estimate.

Ménard is careful about it. Filled sky cannot hold a discovery, and the validation reports come from the same system that built the map, so nothing here has had human peer review yet. What the project does show, and what echoes Anthropic's earlier chemistry work with Claude, is a pattern: an AI assistant absorbing the slow calibration that keeps useful scientific projects from ever shipping.

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Sources:

- Using Claude Science to produce the first complete map of the sky in UV light (Anthropic)
- The full sky in the ultraviolet (map and technical paper, Brice Ménard, Johns Hopkins University)
- Claude Science, an AI workbench for scientists (Anthropic)
- Anthropic's Claude Science creates the first complete ultraviolet map of the sky (The Decoder)
- Ménard works with Claude Science to produce first complete UV light map of the sky (Johns Hopkins Physics & Astronomy)

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